Assignment 2

1. Create a directory named `my_photos`.

```
naman@naman-virtualbox:~$ mkdir my_photos
```

2. Create a subfolder inside `my_photos` named `2024`.

```
naman@naman-virtualbox:~/my_photos$ mkdir 2024
```

3. Create an empty file inside `2024` named `vacation.jpg` (even though it’s just a text placeholder).

```
naman@naman-virtualbox:~/my_photos/2024$ touch vacation.jpg
```

4. Create another empty file in `2024` named `family.png`.

```
naman@naman-virtualbox:~/my_photos/2024$ touch family.png
```

5. List only the files inside `2024` that end with `.png`.

```
naman@naman-virtualbox:~/my_photos/2024$ ls -png
```

6. Copy `family.png` to the `my_photos` directory.

```
naman@naman-virtualbox:~/my_photos/2024$ cp family.png ~/my_photos
```

7. Rename the copy of `family.png` in `my_photos` to `home.png`.

```
naman@naman-virtualbox:~/my_photos$ mv family.png home.png
```

8. Create a directory named `trash`.

```
naman@naman-virtualbox:~$ mkdir trash
```

9. Move `home.png` into the `trash` directory.

```
naman@naman-virtualbox:~$ cd my_photos
naman@naman-virtualbox:~/my_photos$ mv home.png ../trash
```

10. Delete the `trash` directory and everything inside it.

```
naman@naman-virtualbox:~$ rm -r trash
```

11. Create a file named `grocery_list.txt`.

    naman@naman-virtualbox:~$ touch grocery_list.txt

12. Add the word "Milk" to the `grocery_list.txt` file.

    naman@naman-virtualbox:~$ echo "Milk" > grocery_list.txt

13. Add the word "Eggs" to the same file on a new line.

    naman@naman-virtualbox:~$ echo "Eggs" >> grocery_list.txt

14. Display the contents of `grocery_list.txt` on your screen.

    naman@naman-virtualbox:~$ cat grocery_list

    Milk

    Eggs

**15. Count how many lines are in your `grocery_list.txt`.

16. Create a file named `hello.txt` with the text "Hello World".

    naman@naman-virtualbox:~$ echo "Hello World" > hello.txt

17. Overwrite `hello.txt` so it only says "Goodbye".

    naman@naman-virtualbox:~$ echo "Goodbye" > hello.txt

18. Search for the letter "G" in the `hello.txt` file.

    naman@naman-virtualbox:~$ grep "G" hello.txt

19. Create a directory named `scripts`.

    naman@naman-virtualbox:~$ mkdir scripts

20. Move into the `scripts` directory using the `cd` command.

    naman@naman-virtualbox:~$ cd scripts

21. Show the full path of the `scripts` directory you are currently in.

    naman@naman-virtualbox:~$ pwd

22. Create an empty file named `test.py` inside `scripts`.

naman@naman-virtualbox:~/scripts$ touch test.py

23. Move back to your home directory using a single shortcut.

naman@naman-virtualbox:~/scripts$ cd ..

24. List all files in your home directory, including the hidden ones.

naman@naman-virtualbox:~$ la

25. Use the `history` command to see the last 5 commands you typed.

naman@naman-virtualbox:~$ history 5

```
 180  grep "G" hello.txt
 181  la
 182  tail -n 5 history
 183  history
 184  history 5
```

26. Use the `cal` command to see the calendar for the current month.

naman@naman-virtualbox:~$ cal

```
    March 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31
```

27. Find out how long your computer has been powered on (uptime).

naman@naman-virtualbox:~$ uptime

12:07:35 up 3:20, 1 user, load average: 0.07, 0.16, 0.18

28. Clear your terminal screen so it is empty again.

naman@naman-virtualbox:~$ clear

29. Create a file named `sorting.txt` with the numbers 3, 1, and 2 on separate lines.

naman@naman-virtualbox:~$ echo -e "3\n1\n2" > sorting.txt

30. Display the numbers from `sorting.txt` in alphabetical/numerical order.

naman@naman-virtualbox:~$ sort sorting.txt

Shell scripts

1. Write a script that asks for a user's name and prints a "Hello" message.

```
#!/bin/bash

read -p "Enter your name: " name

echo "Hello! $name "
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter your name: Naman
Hello! Naman
```

2. Create a script that takes three separate strings as input and displays them in reverse order.

```
#!/bin/bash

read -p "String1: " str1
```

```
read -p "String2: " str2

read -p "String3: " str3

echo " $str3 $str2 $str1 "
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
String1: I
String2: am
String3: Naman
 Naman am I
```

3. Write a script that asks for a user's age and calculates how many years are left until they turn 100.

```
#!/bin/bash

read -p "Enter your age: " age

LastAge=100

if [ "$age" == "0" ]; then

        echo "Please enter a valid age."

else

        result=$((LastAge-age))

        echo "$result years left to turn 100"

fi
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter your age: 24
76 years left to turn 100
```

4. Create a script that prompts for a filename and then creates that file using `touch`.

```
#!/bin/bash

read -p "Enter the file name " newFile

touch "$newFile"

echo "File created successfully."
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the file name NewFile
File created successfully.
```

5. Write a script that asks for a directory name and lists all files inside that directory.

```
#!/bin/bash

read -p "Enter the directory name: " dir

ls "$dir"
```

```
naman@naman-virtualbox:~/shell_scripts$ mkdir RKJ
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the directory name: RKJ
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the directory name: NV
ls: cannot access 'NV': No such file or directory
naman@naman-virtualbox:~/shell_scripts$
```

6. Write a script to take two numbers as input and display their sum.

```
#!/bin/bash

read -p "Enter first no: " n1

read -p "Enter second no: " n2

sum=$((n1 + n2))

echo "Sum = $sum"
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter first no: 5
Enter second no: 4
Sum = 9
```

7. Create a script that calculates the area of a rectangle (Length × Width) based on user input.

```
#!/bin/bash

read -p "What's the length: " len

read -p "What's the width: " wid

area=$((len*wid))

echo "Rectangle's area: $area"
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
What's the length: 12
What's the width: 10
Rectangle's area: 120
```

8. Write a script that takes a temperature in Celsius and converts it to Fahrenheit.

```
#!/bin/bash

read -p "Enter the temp. in Celsius: " temp

# Formula: °F = (°C × 9/5) + 32

ft=$(((temp * 9/5) + 32))

echo "Temperature in Farenheit is: $ft"
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the temp. in Celsius: 35
Temperature in Farenheit is: 95
```

9. Create a script that performs all four basic operations (+, -, *, /) on two provided numbers.

```
#!/bin/bash

echo "------Welcome to Basic Calculator-----"

read -p "Please select operation: ( + , -, /, * ) " opr

read -p "Please enter 1st number: " n1

read -p "Please enter 2nd number: " n2

if [ "$opr" == "+" ]; then

        result=$((n1+n2))

        echo "Sum is: $result"

elif [ "$opr" == "-" ]; then

        result=$((n1-n2))

        echo "Value is: $result"

elif [ "$opr" == "*" ]; then

        result=$((n1*n2))

        echo "Value is: $result"

elif [ "$opr" == "/" ]; then

        result=$((n1/n2))

        echo "Value is: $result"

else

        echo "Please enter a valid operator"

fi
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
------Welcome to Basic Calculator-----
Please select operation: ( + , -, /, * ) +
Please enter 1st number: 5
Please enter 2nd number: 4
Sum is: 9
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
------Welcome to Basic Calculator-----
Please select operation: ( + , -, /, * ) -
Please enter 1st number: 9
Please enter 2nd number: 7
Value is: 2
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
------Welcome to Basic Calculator-----
Please select operation: ( + , -, /, * ) *
Please enter 1st number: 11
Please enter 2nd number: 7
Value is: 77
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
------Welcome to Basic Calculator-----
Please select operation: ( + , -, /, * ) /
Please enter 1st number: 6
Please enter 2nd number: 3
Value is: 2
```

10. Write a script to calculate the square and cube of a number entered by the user.

```
#!/bin/bash

read -p "Enter the number: " num

sq=$((num*num))

cube=$((sq*num))

echo "Square of the number is: $sq and Cube of the number is: $cube"
```

```
naman@naman-virtualbox:~/shell_scripts$ nano test.sh
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the number: 39
Square of the number is: 1521 and Cube of the number is: 59319
```

11. Create a script that finds the remainder (modulus) when one number is divided by another.

```
#!/bin/bash

read -p "Enter first no: " n1

read -p "Enter second no: " n2

mod=$((n1%n2))

echo "Remainder is: $mod"
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter first no: 87
Enter second no: 13
Remainder is: 9
```

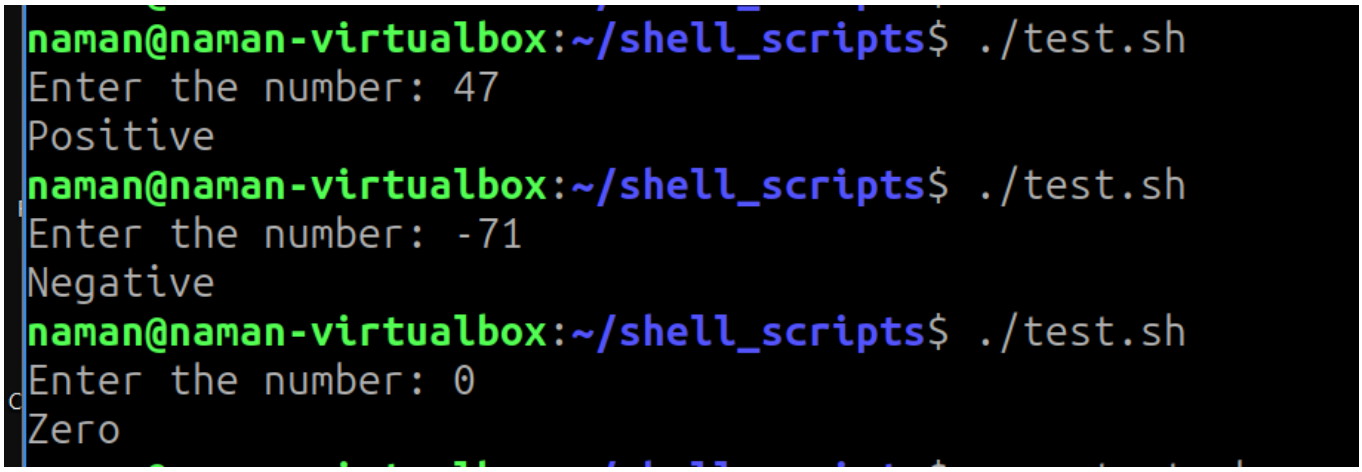
12. Write a script that takes a total number of minutes and converts it into "X hours and Y minutes" format.

```
#!/bin/bash

read -p "Enter the no. of minutes: " min

hr=$((min/60))

mins=$((min%60))

echo "--> $hr Hours and $mins minutes"
```

```
naman@naman-virtualbox:~/shell_scripts$ nano test.sh
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the no. of minutes: 144
--> 2 Hours and 24 minutes
```

13. Write a script that checks if a number entered by the user is positive, negative, or zero.

```
#!/bin/bash

read -p "Enter the number: " num

if (( $num > 0 )); then

        echo "Positive"

elif (( $num < 0 )); then

        echo "Negative"

else

        echo "Zero"

fi
```



14. Create a script that asks for a number and determines if it is even or odd.

```
#!/bin/bash

read -p "Enter a number: " num

if (( $num % 2 == 0 )); then

        echo "It's Even"

else
```

```
    echo "It's Odd"
fi
```

```
naman@naman-virtualbox:~/shell_scripts$ nano test.sh
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter a number: 12
It's Even
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter a number: 79
It's Odd
```

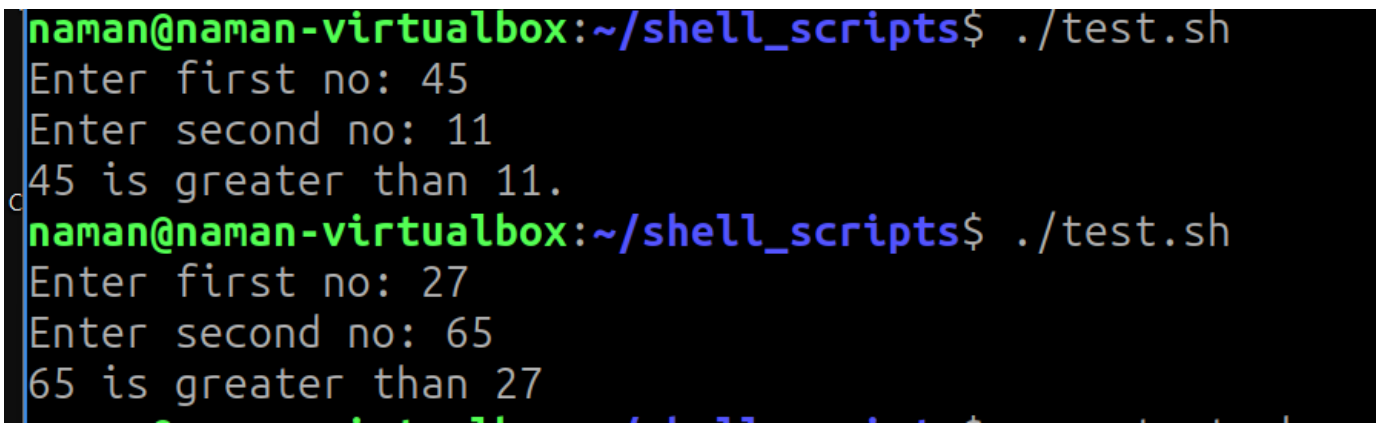
15. Write a script that checks if a user is eligible to vote (age 18 or older).

```
#!/bin/bash
read -p "Please enter your age: " age
if (( $age >= 18 )); then
    echo "Eligible"
else
    echo "Not Eligible"
fi
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Please enter your age: 23
Eligible
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Please enter your age: 17
Not Eligible
```

16. Create a script that compares two numbers and prints which one is larger.

```
#!/bin/bash

read -p "Enter first no: " n1

read -p "Enter second no: " n2

if (( $n1 > $n2 )); then

        echo "$n1 is greater than $n2."

elif (( $n1 < $n2 )); then

        echo "$n2 is greater than $n1"

else

        echo "Both are equal"

fi
```



17. Write a "Pass/Fail" script: if a score is 50 or above, print "Pass", otherwise "Fail".

```
#!/bin/bash

read -p "Enter the marks: " marks


if (( $marks >= 50 )); then

        echo "Pass"

else
```

```
        echo "Fail"
fi
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the marks: 97
Pass
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the marks: 32
Fail
```

18. Create a script that checks if a specific file (input by the user) exists on the system.

```
#!/bin/bash

read -p "Enter the file name: " path

if [ -f "$path" ]; then

        echo "It is a file"

else

        echo "Not a file"

fi
```

```
naman@naman-virtualbox:~/shell_scripts$ touch atulsir.txt
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the file name: atulsir.txt
It is a file
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the file name: siratul.txt
Not a file
```

19. Write a script that asks for a password and prints "Access Granted" if it matches "Secret123", and "Access Denied" otherwise.

```
#!/bin/bash

read -p "Enter password: " pass
```

```
if [ "$pass" == "Secret123" ]; then
        echo "Access Granted"
else
        echo "Access Denied"
fi
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter password: Secret123
Access Granted
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter password: secret648'
Access Denied
```

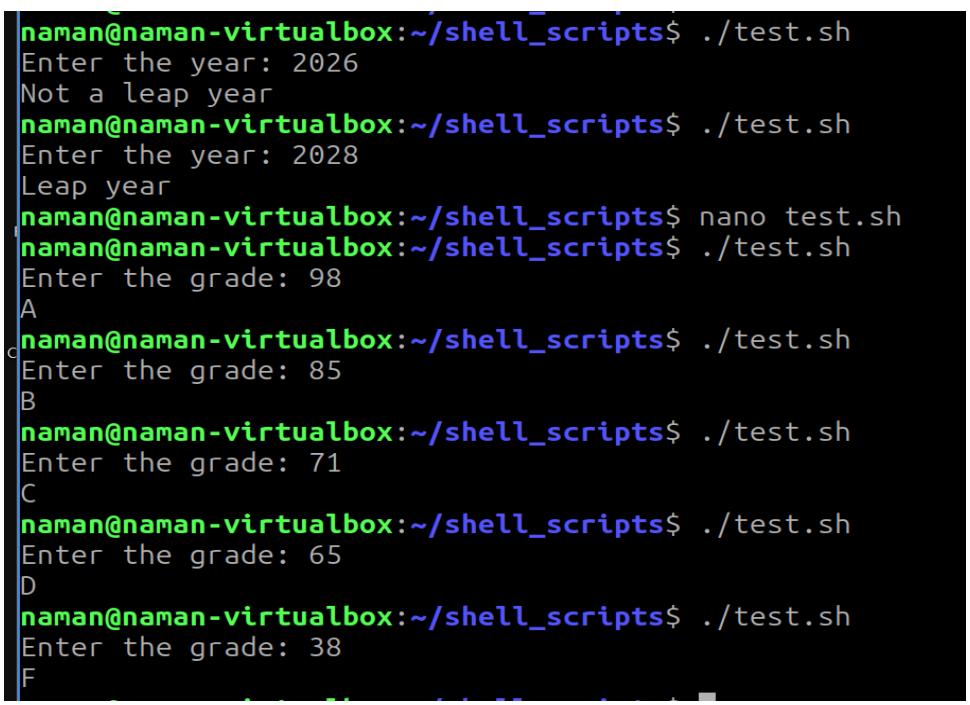
20. Create a script that checks if a given year is a leap year.

```
#!/bin/bash
read -p "Enter the year: " year
if (( $year % 4 == 0 )); then
        echo "Leap year"
else
        echo "Not a leap year"
fi
```

```
naman@naman-virtualbox:~/shell_scripts$ nano test.sh
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the year: 2026
Not a leap year
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the year: 2028
Leap year
```

21. Write a script that takes a numerical grade (0-100) and prints the letter grade (A, B, C, D, or F) using `elif`.

```
#!/bin/bash

read -p "Enter the grade: " grade

if (( $grade >= 90 )); then

        echo "A"

elif (( $grade >= 80 )); then

        echo "B"

elif (( $grade >= 70 )); then

        echo "C"

elif (( $grade >= 50 )); then

        echo "D"

else

        echo "F"

fi
```



22. Create a script that asks for a username; if the username is "admin", it asks for a password; otherwise, it exits.

```
#!/bin/bash

read -p "Enter the username: " user

if [  "$user" == "admin" ]; then

        read -p "Please enter password: " pass

fi
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the username: admin
Please enter password: 123
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the username: atulsir
```

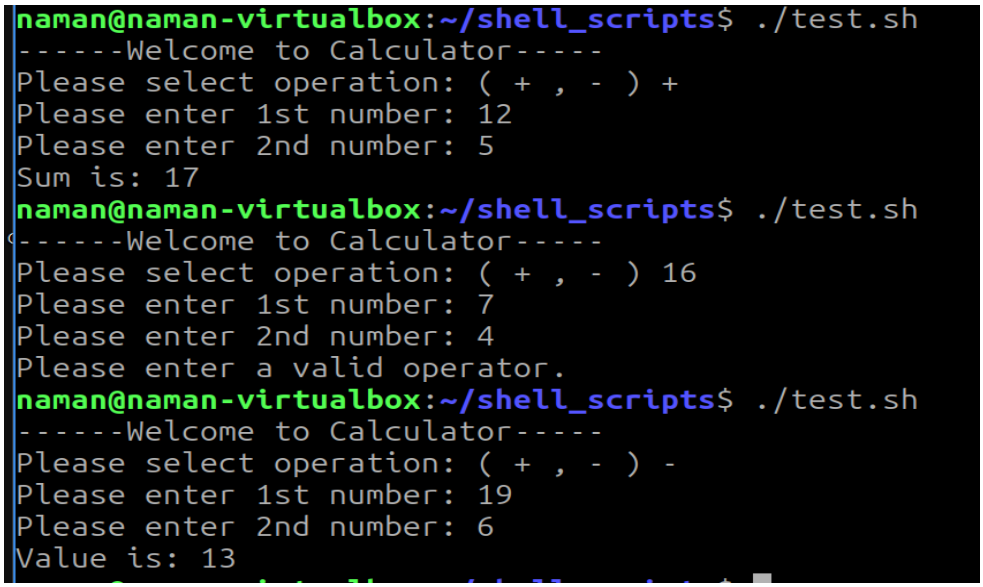
23. Write a script that checks if a string entered by the user is empty or not.

```
#!/bin/bash

read -p "Enter: " word

if [ -z "$word" ]; then

      echo "Empty"

else

      echo "Not empty"

fi
```

```
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter: atulsir
Not empty
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter:
Empty
```

24. Create a simple calculator where the user enters two numbers and an operator (+ or -) to get the result.

```
#!/bin/bash

echo "------Welcome to Calculator-----"

read -p "Please select operation: ( + , - ) " opr

read -p "Please enter 1st number: " n1

read -p "Please enter 2nd number: " n2

if [ "$opr" == "+" ]; then

        result=$((n1+n2))

        echo "Sum is: $result"

elif [ "$opr" == "-" ]; then

        result=$((n1-n2))

        echo "Value is: $result"

else

        echo "Please enter a valid operator."

fi
```



25. Write a script that checks if a user-provided path is a file or a directory.

```
#!/bin/bash

read -p "Enter the file/directory name: " path

if [ -f "$path" ]; then

        echo "It is a file"

elif [ -d "$path" ]; then

        echo "It is a directory"

else

        echo "File/Directory doesn't exist."

fi
```

```
naman@naman-virtualbox:~/shell_scripts$ ls
atulsir.txt  NewFile  RKJ  test.sh
naman@naman-virtualbox:~/shell_scripts$ nano test.sh
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the file/directory name: atulsir.txt
It is a file
naman@naman-virtualbox:~/shell_scripts$ ./test.sh
Enter the file/directory name: RKJ
It is a directory
```